

Rounding switching controls under a hard switch budget: sharp minimax bounds and exact algorithms

Abstract

Combinatorial integral approximation replaces a relaxed mixture of control modes by one active mode at a time while controlling cumulative error. A hard bound on the number of switches changes the approximation problem: even on arbitrarily fine time grids, the worst-case error can remain proportional to the horizon. We study this error as a function of the number of modes, the switch budget, and the time grid. We establish exact uniform-input formulas and, for at least three modes, the full continuous one-switch minimax value, with matching extremizers and constructive proofs for measurable relaxed controls. These results give explicit counterexamples to a published switching-budget conjecture. The continuous and grid problems are formulated separately, so that bounds for individual inputs and worst-case bounds have unambiguous meanings.

1 Problem and basic properties

A relaxed switching control allocates one unit of activity among finitely many modes. Rounding selects one mode at a time. The cumulative allocation error is the objective of combinatorial integral approximation (CIA), the rounding subproblem in a decomposition of mixed-integer optimal control. A restriction on the number of mode changes is a hard constraint on that subproblem. Sager and Zeile [1] study this formulation, develop maximum dwell rounding, and give sharp bounds and a conjecture for its worst-case error. Our focus is the extremal rounding problem itself. In particular, a bound on cumulative allocation error is not an assertion that the same schedule optimizes an underlying nonlinear control objective.

1.1 Controls, schedules, and the two error criteria

Fix a horizon $T > 0$ and an integer number of modes $n \geq 2$. Throughout, switch budgets s are nonnegative integers, activation-block counts k are positive integers, and grid sizes N are positive integers. Write $[n] = \{1, \dots, n\}$ and

$$\Delta_n = \left\{ a \in \mathbb{R}^n : a_i \geq 0, \sum_{i=1}^n a_i = 1 \right\}.$$

A relaxed control is a Lebesgue-measurable map $\alpha : [0, T] \rightarrow \Delta_n$, considered up to equality almost everywhere. Its cumulative allocation and terminal masses are

$$A_i(t) = \int_0^t \alpha_i(u) du, \quad m_i = A_i(T). \quad (1)$$

Thus A_i is nondecreasing and 1-Lipschitz, $A_i(0) = 0$, and $\sum_i A_i(t) = t$. Conversely, every vector of functions with these properties is the cumulative allocation of such a control:

Lipschitz functions are absolutely continuous and differentiable almost everywhere, their derivatives are nonnegative, and differentiation of the sum gives a simplex-valued vector almost everywhere. Denote this class of cumulative allocations by $\mathcal{A}_n(T)$. We use either α or A to specify an input.

An integer control, or schedule, selects a coordinate vector e_i on each of finitely many intervals. Its initial activation is free: only changes between two successive active modes count as switches. Its values at isolated times are immaterial. Let $\mathcal{W}_{n,k}(T)$ be the set of cumulative occupations

$$W_i(t) = \int_0^t \omega_i(u) du$$

of schedules with at most k positive-length activation blocks, equivalently at most $s = k - 1$ switches. Unless explicitly stated otherwise, repeated modes are allowed, the initial and final modes are unrestricted, and there are no dwell-time or transition restrictions. A parameterization of this entire set uses a word $p \in [n]^k$ and ordered times

$$0 = t_0 \leq t_1 \leq \dots \leq t_k = T, \quad \omega(t) = e_{p_j} \quad (t_{j-1} < t < t_j). \quad (2)$$

Zero-length intervals can be deleted and consecutive equal modes can be merged. These operations cannot increase the switch count. Consequently this parameterization includes shorter schedules without imposing distinctness on competing modes.

The full error and the one-sided error are respectively

$$D(A, W) = \max_{i \in [n]} \max_{0 \leq t \leq T} |A_i(t) - W_i(t)|, \quad D^-(A, W) = \max_{i \in [n]} \max_{0 \leq t \leq T} (W_i(t) - A_i(t)). \quad (3)$$

The minus superscript refers to the negative part of the discrepancy $A - W$; the displayed one-sided maximum is nonnegative because it includes $t = 0$. For a fixed input, define the continuous instance optima

$$\text{OPT}_s(A) = \min_{W \in \mathcal{W}_{n,s+1}(T)} D(A, W), \quad \text{OPT}_k^-(A) = \min_{W \in \mathcal{W}_{n,k}(T)} D^-(A, W). \quad (4)$$

The corresponding minimax values are

$$F_{n,s}(T) = \sup_{A \in \mathcal{A}_n(T)} \text{OPT}_s(A), \quad G_{n,k}^-(T) = \sup_{A \in \mathcal{A}_n(T)} \text{OPT}_k^-(A). \quad (5)$$

The indexing is intentional: F counts switches, whereas G^- counts activation blocks, with $k = s + 1$. Both infima in (4) and both suprema in (5) are attained, as shown below.

1.2 Fixed grids and endpoint evaluation

A grid is a strictly increasing sequence $\mathcal{T} = (x_0, \dots, x_N)$ with $x_0 = 0$, $x_N = T$. Its cell lengths are $\Delta_j = x_j - x_{j-1}$ and $\bar{\Delta} = \max_j \Delta_j$. Let $\mathcal{A}_n(\mathcal{T})$ consist of cumulative allocations of controls constant on each cell, and let $\mathcal{W}_{n,k}(\mathcal{T})$ restrict the switching times to grid points. Set

$$\text{OPT}_s^{\mathcal{T}}(A) = \min_{W \in \mathcal{W}_{n,s+1}(\mathcal{T})} D(A, W), \quad F_{n,s}^{\mathcal{T}} = \max_{A \in \mathcal{A}_n(\mathcal{T})} \text{OPT}_s^{\mathcal{T}}(A). \quad (6)$$

The instance optimum is defined even for $A \in \mathcal{A}_n(T)$ that is not piecewise affine. The grid minimax in (6) does restrict the input to cellwise constant controls. Analogously, use $\text{OPT}_k^{-,\mathcal{T}}$ and $G_{n,k}^{-,\mathcal{T}}$ for the one-sided criterion. On the unit grid $(0, 1, \dots, N)$ we abbreviate the full minimax by $F_{n,s}^{\text{unit}}(N)$. Every grid with cell length Δ is obtained from a unit grid by scaling all errors by Δ .

For the same input, $\text{OPT}_s(A) \leq \text{OPT}_s^{\mathcal{T}}(A)$. This fact alone does not compare the two full minimax values, whose input classes differ. The following observation permits exact evaluation even for merely measurable inputs and will be used throughout.

Lemma 1.1 (Endpoint monotonicity). *On any interval on which the selected mode is p , the discrepancy $A_p - W_p$ is nonincreasing and every other discrepancy $A_i - W_i$ is nondecreasing. Hence both errors in (3) are determined by the activation-block endpoints. In particular, for a grid schedule they are determined by grid endpoints, even when α varies inside grid cells.*

Proof. The absolutely continuous discrepancies have derivatives $\alpha_p - 1 \leq 0$ and $\alpha_i \geq 0$ for $i \neq p$, almost everywhere on the interval. Integrating these inequalities proves monotonicity. A monotone real function takes its largest absolute value and each of its one-sided extrema at an endpoint. $\square$

Replacing a measurable control by its cell averages preserves every grid endpoint $A_i(x_j)$. Lemma 1.1 therefore shows that this replacement preserves the error of every grid schedule, and hence its grid instance optimum. This observation does not say that averaging preserves its continuous instance optimum. It does, however, give the minimax comparison

$$F_{n,s}(T) \leq F_{n,s}^T, \quad G_{n,k}^-(T) \leq G_{n,k}^{-,T}. \quad (7)$$

Indeed, for any measurable input A , its cell average $\bar{A}$ satisfies $\text{OPT}_s(A) \leq \text{OPT}_s^T(A) = \text{OPT}_s^T(\bar{A}) \leq F_{n,s}^T$. Taking the supremum proves the first inequality; the one-sided argument is identical.

1.3 Existence, continuity, and scaling

Equip vector-valued cumulative functions with $\|A\|_\infty = \max_i \max_{t \in [0,T]} |A_i(t)|$.

Proposition 1.2 (Attainment and stability). *The sets $\mathcal{A}_n(T)$ and $\mathcal{W}_{n,k}(T)$ are compact in the uniform norm. Each of the four instance optimum maps defined above is 1-Lipschitz in A . All continuous and grid instance optima and minimax values are attained.*

Proof. The cumulative allocations are uniformly bounded and equicontinuous. Their uniform limits preserve monotonicity, the Lipschitz constants, initial values, and the sum identity, and therefore belong to $\mathcal{A}_n(T)$ by the converse following (1). The Arzelà–Ascoli theorem gives compactness.

For a fixed word p , the ordered time simplex in (2) is compact. Its cumulative occupations are continuous in those times. Indeed, the useful representation

$$W_i(t) = \mathbf{1}_{\{p_k=i\}}t + \sum_{j=1}^{k-1} (\mathbf{1}_{\{p_j=i\}} - \mathbf{1}_{\{p_{j+1}=i\}}) \min\{t, t_j\}$$

gives $\|W(\cdot; t_1, \dots, t_{k-1}) - W(\cdot; t'_1, \dots, t'_{k-1})\|_\infty \leq \sum_{j=1}^{k-1} |t_j - t'_j|$. Taking the finite union over words proves compactness of $\mathcal{W}_{n,k}(T)$. Both error criteria are continuous, so their minima are attained. For either criterion d , the inequality $|d(A, W) - d(B, W)| \leq \|A - B\|_\infty$ holds for every W . Taking minima shows the stated Lipschitz property. Compactness of $\mathcal{A}_n(T)$ then gives attainment of the outer maxima. For grids, there are finitely many schedules and the input is a product of N compact simplexes, proving the analogous claims. $\square$

Changing time by $t = Tu$ is a bijection of both input and schedule classes and multiplies errors by T . Thus

$$F_{n,s}(T) = TF_{n,s}(1), \quad G_{n,k}^-(T) = TG_{n,k}^-(1). \quad (8)$$

The same statement applies to instance optima and a simultaneously rescaled grid. At $T = 0$ all errors are zero; results stated for $T > 0$ extend with this convention. Enlarging the switch budget cannot increase any instance optimum or minimax value.

Proposition 1.3 (No-switch baseline). *For every $n \geq 2$, $F_{n,0}(T) = T(1 - 1/n)$. On any grid, $F_{n,0}^T = T(1 - 1/n)$ as well.*

Proof. Activating mode p throughout gives error $\max\{T - m_p, \max_{i \neq p} m_i\} = T - m_p$, since $\sum_{i \neq p} m_i = T - m_p$. Choosing a largest terminal mass gives error at most $T(1 - 1/n)$. The uniform control $\alpha_i = 1/n$ gives equality and belongs to every grid input class. $\square$

2 Uniform inputs and the exact one-switch minimax

We first solve a benchmark family for every budget with fewer activation blocks than modes. This family gives a dimension-dependent obstruction to small error. We then prove that its obstruction, together with an omitted-mode obstruction, determines the full continuous one-switch minimax for at least three modes.

2.1 Exact uniform-input error in continuous time

Write $A_i^{\text{unif}}(t) = t/n$ and let $E_{n,s}(T) = \text{OPT}_s(A^{\text{unif}})$.

Theorem 2.1 (Uniform controls). *Let $n \geq 2$ and $0 \leq s \leq n - 2$. With $k = s + 1$ and $r = n/(n - 1)$,*

$$E_{n,s}(T) = T \max \left\{ \frac{1}{n}, \frac{1}{n(r^k - 1)} \right\}. \quad (9)$$

A schedule using k distinct modes attains this value, although competing schedules may repeat modes.

Proof. Consider any schedule with $m \leq k < n$ positive-length blocks and error E . At least one mode is unused, so $E \geq T/n$. Let $0 = t_0 < \dots < t_m = T$ be its block endpoints. At the end of block j , the occupation of its active mode p_j is at least $t_j - t_{j-1}$, even if p_j occurred earlier. The inequality $W_{p_j}(t_j) - t_j/n \leq E$ implies

$$t_j \leq r t_{j-1} + r E. \quad (10)$$

Iterating from zero yields $T \leq nE(r^m - 1) \leq nE(r^k - 1)$ and proves both lower bounds.

For attainment, set

$$E_0 = \frac{T}{n(r^k - 1)}, \quad t_j = T \frac{r^j - 1}{r^k - 1} \quad (0 \leq j \leq k),$$

and choose a distinct mode on each block. Before its activation, the selected mode has discrepancy t/n ; during activation its discrepancy decreases; after activation it increases again. At its block endpoint,

$$A_{p_j}(t_j) - W_{p_j}(t_j) = \frac{t_j}{n} - (t_j - t_{j-1}) = -E_0.$$

Every positive discrepancy is at most T/n , and no selected mode's negative discrepancy exceeds E_0 . Unused modes also have error at most T/n . This proves attainment of (9). $\square$

For one switch, this gives

$$E_{n,1}(T) = T \max \left\{ \frac{1}{n}, \frac{(n-1)^2}{n(2n-1)} \right\} \quad (n \geq 3). \quad (11)$$

For every fixed $s \geq 0$, the uniform value tends to $T/(s+1)$ as $n \rightarrow \infty$. Indeed, with fixed $k = s+1$, $(1 + 1/(n-1))^k - 1 = k/(n-1) + O_k(n^{-2})$, so $n(r^k - 1) \rightarrow k$. This is an obstruction supplied by one input family; it does not presume that this family maximizes the error over all inputs.

2.2 A scalar recurrence on uniform grids

Let $E_{n,s}^{\text{unit}}(N)$ be the grid instance optimum for the uniform control on N unit cells.

Theorem 2.2 (Exact integer recurrence). *Under $n \geq 2$, $0 \leq s \leq n-2$, and $N \geq 1$, define, for integers $K \geq N$,*

$$b_0(K) = 0, \quad b_{j+1}(K) = \left\lfloor \frac{nb_j(K) + K}{n-1} \right\rfloor. \quad (12)$$

If K_ is the least such integer for which $b_{s+1}(K_*) \geq N$, then $E_{n,s}^{\text{unit}}(N) = K_*/n$. Moreover,*

$$E_{n,s}(N) \leq E_{n,s}^{\text{unit}}(N) < E_{n,s}(N) + \frac{n-1}{n}. \quad (13)$$

Proof. At integer times, every discrepancy is a multiple of $1/n$; by Lemma 1.1 an optimal error is therefore K/n for an integer K . Every admissible schedule omits a mode, giving $K \geq N$. The block-end argument (10), with integer block endpoints, gives $t_j \leq \lfloor (nt_{j-1} + K)/(n-1) \rfloor$. Monotonicity of this recurrence implies $t_j \leq b_j(K)$ and hence necessity of $b_{s+1}(K) \geq N$. Fewer blocks can be padded by zero-length blocks; alternatively the nondecreasing iterates give the same conclusion.

Conversely, suppose the condition holds. Put $t_j = \min\{N, b_j(K)\}$ and stop when N is first reached. Since the recurrence cannot leave zero if $b_1(K) = 0$, the assumed condition implies $b_1(K) \geq 1$, or $K \geq n-1$. It follows that the iterates strictly increase before truncation. Assign a distinct mode to each positive-length block. Its block-end inequality gives $W_i(t_j) - t_j/n \leq K/n$. Before and after that block, its discrepancy is nondecreasing, so this bounds its negative part throughout. Positive errors of selected and unused modes are at most $N/n \leq K/n$. There are at most $s+1$ blocks, proving sufficiency.

The predicate is monotone in K , and $K = N(n-1)$ already reaches N in one step. Thus K_* exists and binary search may use $N \leq K \leq N(n-1)$, with $s+1$ recurrence evaluations per trial. To prove the strict gap, write $E = E_{n,s}(N)$ and $C = \lceil nE \rceil$, and test $K = C + n - 2 \geq N$. For every integer b ,

$$\left\lfloor \frac{nb + C + n - 2}{n-1} \right\rfloor = \left\lceil \frac{nb + C}{n-1} \right\rceil \geq rb + rE.$$

The discrete recurrence therefore dominates the continuous recurrence starting at zero, which reaches at least N in $s+1$ steps by (9). Consequently $E_{n,s}^{\text{unit}}(N) \leq K/n < E + (n-1)/n$. The lower inequality follows by restricting continuous switching times to integer times for the same input. $\square$

2.3 The continuous one-switch minimax

For two distinct modes p, q , let $W^{p,q,\tau}$ activate p before time τ and q afterward, allowing $\tau = 0$ or T . These endpoints include constant schedules.

Lemma 2.3 (The three-term formula). *For $n \geq 2$ and $0 \leq \tau \leq T$, taking the maximum over omitted terminal masses as zero when $n = 2$,*

$$D(A, W^{p,q,\tau}) = \max \left\{ \max_{i \notin \{p,q\}} m_i, \quad \tau - A_p(\tau), \quad T - m_q - \tau \right\}. \quad (14)$$

Proof. The omitted modes increase to their terminal masses. The discrepancy of p decreases from zero to $A_p(\tau) - \tau$ and then increases to $m_p - \tau$. Its possible positive final error is

bounded by $T - m_q - \tau$, because $m_p + m_q \leq T$. The discrepancy of q increases to $A_q(\tau)$ and then decreases to $m_q - (T - \tau)$. Its positive error at activation is at most $\tau - A_p(\tau)$, because $A_p(\tau) + A_q(\tau) \leq \tau$. These comparisons bound every endpoint extremum by the right-hand side. Conversely, the first two displayed terms are actual errors. If the last term is positive it is an actual negative terminal discrepancy; if it is nonpositive it is dominated by the nonnegative second term. This proves the equality, including $\tau = 0, T$. $\square$

Theorem 2.4 (Exact continuous one-switch minimax). *For every $n \geq 3$,*

$$F_{n,1}(T) = H_n(T) := T \max \left\{ \frac{1}{3}, \frac{(n-1)^2}{n(2n-1)} \right\}. \quad (15)$$

For $n = 3, 4$, three consecutive distinct pure modes, each of duration $T/3$, attain the maximum. For $n \geq 5$, the uniform control attains it. A schedule with the asserted error bound can be constructed using terminal masses and cumulative allocations at at most two additional times.

Proof. By scaling take $T = 1$ and put $E = H_n(1)$. Directly from the formula, $1/3 \leq E < 1/2$.

First suppose some $m_q \geq E$. At most one other mode has mass strictly larger than E : two such modes together with q would have total exceeding $3E \geq 1$. Choose $p \neq q$ to include that mode if it exists, and choose any other mode otherwise. All omitted masses are at most E . With $\tau = \max\{0, 1 - m_q - E\}$ we have $0 \leq \tau \leq \max\{0, 1 - 2E\} \leq E$. Thus $\tau - A_p(\tau) \leq E$ and $1 - m_q - \tau \leq E$, and Lemma 2.3 gives a feasible schedule.

It remains to treat the case $m_i < E$ for every mode. Let q index a largest mass and r a second largest, breaking ties arbitrarily. Define

$$t_q = 1 - m_q - E, \quad t_r = 1 - m_r - E, \quad h_i = 1 - m_i - 2E.$$

Since $m_i < E < 1/2$, the times satisfy $0 < t_q \leq t_r < 1$. Consider all schedules $p \rightarrow q$, $p \neq q$, at time t_q , and also $q \rightarrow r$ at time t_r . Their omitted masses are below E , and their last terms in (14) equal E . If all these schedules had error strictly larger than E , the middle terms would force

$$A_p(t_q) < h_q \quad (p \neq q), \quad A_q(t_q) \leq A_q(t_r) < h_r.$$

Summing gives $t_q < (n-1)h_q + h_r$, or

$$(2n-1)E < n-1 - (n-2)m_q - m_r. \quad (16)$$

But $m_q \geq 1/n$ and $m_r \geq (1 - m_q)/(n-1)$. For $n \geq 3$ the coefficient $n-2-1/(n-1)$ is positive, so

$$(n-2)m_q + m_r \geq \left(n-2 - \frac{1}{n-1} \right) m_q + \frac{1}{n-1} \geq \frac{n-1}{n}.$$

Equation (16) would imply $(2n-1)E < (n-1)^2/n$, contrary to the definition of E . At least one candidate therefore has error at most E .

For a matching lower bound of $1/3$, use three consecutive distinct pure modes of duration $1/3$. Every schedule with at most two blocks omits at least one of these modes and incurs its terminal mass. The uniform-input value in (11) supplies the other lower bound. Finally,

$$\frac{(n-1)^2}{n(2n-1)} - \frac{1}{3} = \frac{n^2 - 5n + 3}{3n(2n-1)}$$

is negative at $n = 3, 4$ and positive for every integer $n \geq 5$, giving the claimed transition and extremizers. $\square$

For an arbitrary horizon $T > 0$, set $E = H_n(T)$ and $t_i = T - m_i - E$. The construction requires no search over switching times. Let q and r index a largest and second-largest terminal mass. If $m_q \geq E$, select $r \rightarrow q$ at $\max\{0, T - m_q - E\}$. Otherwise evaluate $A(t_q)$ and choose $p \neq q$ maximizing $A_p(t_q)$. If its candidate fails, the $q \rightarrow r$ candidate at t_r succeeds. Apart from cumulative integration, this requires $O(n)$ arithmetic operations and comparisons. The proof covers ties, zero terminal masses, and switching at an endpoint.

Corollary 2.5 (One-switch grid upper bound). *For every grid $\mathcal{T}$ and every measurable relaxed input,*

$$\text{OPT}_1^{\mathcal{T}}(A) \leq H_n(T) + \frac{\bar{\Delta}}{2} \quad (n \geq 3).$$

The coefficient $H_n(T)/T$ cannot be decreased in a bound whose grid term vanishes under refinement, uniformly over grids and inputs.

Proof. Choose the schedule from Theorem 2.4 and move its switch to a closest grid point. The displacement is at most $\bar{\Delta}/2$. The cumulative occupation of either affected mode changes by at most that displacement, so the triangle inequality gives the bound. No new switch is created. For sharpness of the leading coefficient, use the uniform extremizer when $n \geq 5$. When $n = 3, 4$, use the three pure blocks on successively refined grids containing $T/3$ and $2T/3$. Each input belongs to its grid class, and its grid optimum is at least its continuous optimum $H_n(T)$. $\square$

2.4 Exact three-cell minimax and a published conjecture

Proposition 2.6 (Three unit cells). *For $n \geq 3$, $F_{n,1}^{\text{unit}}(3) = 2 - 3/n$.*

Proof. For arbitrary relaxed input choose q with a largest terminal mass and p with a second largest, and use the three-cell word (p, q, q) . Every omitted mass is at most 1, because the masses sum to 3. The first-cell error is at most 1. At time 2, the selected modes have occupation 1 and cumulative allocations in $[0, 2]$, hence errors at most 1; omitted modes still have errors at most 1. At time 3, $m_p \leq 3/2$ and its occupation is 1, so its error is at most 1. The final mode satisfies $3/n \leq m_q \leq 3$ and has occupation 2, giving error at most $\max\{1, 2 - 3/n\} = 2 - 3/n$. Endpoint monotonicity proves the upper bound. For uniform input, every schedule with at most one switch gives some mode at least two cells. Its negative terminal discrepancy is at least $2 - 3/n$. Thus the upper bound is attained. $\square$

Sager and Zeile [1, Conjecture 1, equation (7.6)] propose, for $n > 2$ and $1 \leq s \leq N - 2$, the worst-case grid value

$$\begin{cases} T/(s+2) + \bar{\Delta}/2, & s \leq n-2, \\ T/(2s+4-n) + \bar{\Delta}/2, & s > n-2. \end{cases} \quad (17)$$

Their initial activation is free, as in our formulation [1, Definition 10]. The following examples satisfy all of the stated restrictions and disprove the first branch even as an upper bound. No conclusion about every case of its second branch is needed here.

Example 2.7 (Five modes and one switch). Take $n = 5$, $T = N = 45$ unit cells, and $\alpha_i = 1/5$ throughout. For any distinct pair of modes and an integer switch time τ , the exact error from (14) is

$$D(\tau) = \max\{9, 4\tau/5, 36 - \tau\}.$$

The two nonconstant terms intersect at $\tau = 20$, with value 16. For $\tau \leq 20$ the last term is at least 16; for $\tau \geq 20$ the middle term is at least 16. Constant schedules are included by

the endpoint choices. Thus the grid optimum is 16, whereas (17) gives $45/3 + 1/2 = 31/2$. On $45m$ unit cells the same construction has optimum $16m$, exceeding the conjectured $15m + 1/2$ for every integer $m \geq 1$. Equivalently, on a fixed horizon T with $45m$ equal cells its optimum is $16T/45$, while the conjectured bound tends to $T/3$. The failure persists under refinement.

Example 2.8 (Smallest admissible number of cells). On three unit cells and with seven uniform modes, the exact grid error is $11/7$ by Proposition 2.6. It exceeds the conjectured $3/3 + 1/2 = 3/2$. The restriction $1 \leq s \leq N - 2$ excludes grids with fewer than three cells, so this witness uses the smallest admissible number of cells.

More generally, for every fixed $s \geq 1$, Theorem 2.1 approaches $T/(s+1) > T/(s+2)$ as the mode count grows. For sufficiently many modes, its continuous uniform value already exceeds the first branch's proposed leading term. A sufficiently fine uniform grid then makes $\bar{\Delta}/2$ smaller than this strict excess; restricting the uniform input to grid schedules cannot reduce its optimum. Thus this obstruction occurs for every fixed positive switch budget, not only for one switch. In the one-switch case, Theorem 2.4 and Corollary 2.5 give an exact continuous replacement and a valid grid upper bound. The latter is not asserted to be an exact finite-grid equality.

3 Heavy modes and the reduction to one-sided error

The two discrepancy signs have different obstructions. An omitted mode contributes its entire terminal mass to $A - W$, whereas a long activation block can make $W - A$ large. The following theorem separates these effects for every block count. A mode is called *heavy at threshold E* when $m_i > E$; the inequality is strict.

Theorem 3.1 (Universal heavy-mode rounding). *Let $s \geq 0$, $E > 0$, and $T \leq (s+2)E$. If some mode has $m_i > E$, then there is a schedule with at most s switches and full error at most E . There is no restriction relating n to s .*

We prove the theorem by rounding prefixes on equal intervals and then reordering a carefully chosen prefix. This reordering is needed because a repeated mode alone does not save a switch: its occurrences must be adjacent.

Lemma 3.2 (Integral prefix rounding with a repetition). *Let $A \in \mathcal{A}_n(M)$, where $M \geq 2$ is an integer. If $A_h(M) > 1$, there is a word $w_1, \dots, w_M$ on unit intervals whose counts $N_i(j)$ obey*

$$\lfloor A_i(j) \rfloor \leq N_i(j) \leq \lceil A_i(j) \rceil \quad (i \in [n], 1 \leq j \leq M), \quad (18)$$

and in which mode h occurs at least twice. This word has full error at most one for the original measurable input.

Proof. Build a finite network with one supply-one node for each slot j and, for each mode i , a chain of nodes $(i, 1), \dots, (i, M)$. Slot j has an arc to each (i, j) , with capacity one. The chain edge from (i, j) to $(i, j+1)$ has lower and upper capacities $\lfloor A_i(j) \rfloor$ and $\lceil A_i(j) \rceil$; the last chain edge goes to a common sink with the corresponding capacities at M . The sink has demand M . Sending $A_i(j) - A_i(j-1)$ from slot j to its mode chain gives a feasible fractional flow. All bounds and supplies are integers. The integral-flow theorem (equivalently, total unimodularity of a directed incidence matrix) gives an integral optimum for a linear objective on this bounded feasible flow polytope.

Maximize the final flow in chain h . Its optimum is at least the feasible fractional value $A_h(M) > 1$, so an integral optimum assigns at least two slots to h . Each slot sends its one

unit to exactly one mode, and its chain counts satisfy (18). The endpoint error is at most one. Lemma 1.1 extends this bound throughout each unit interval without any assumption that the input is constant there. $\square$

Lemma 3.3 (First-repeat reordering). *Any word on M unit intervals with full error at most one and a repeated mode can be replaced by a word with full error at most one and an equal adjacent pair.*

Proof. If there is already an adjacent pair, nothing is required. Otherwise let r be the first position repeating an earlier mode q . The prefix of length r contains q twice and $r - 2$ other modes once each. Its error at r gives

$$1 \leq A_q(r) \leq 3, \quad A_i(r) \leq 2 \text{ for other selected modes}, \quad A_i(r) \leq 1 \text{ for omitted modes.} \quad (19)$$

Only this prefix will be reordered, preserving its occupation multiset.

Let $d_q = \max\{t \in [0, r] : A_q(t) \leq 1\}$. Continuity and (19) give $A_q(d_q) = 1$, including the case $A_q(r) = 1$ where $d_q = r$. Moreover $1 \leq d_q \leq r$, since $A_q(t) \leq t$. Choose an integer $j \in \{0, \dots, r - 2\}$ with $j \leq d_q \leq j + 2$; one choice is $j = \min\{r - 2, \lceil d_q \rceil - 2\}$. Place q on the double block $[j, j + 2]$.

For another selected mode i with $A_i(r) > 1$, define its deadline $d_i = \max\{t \leq r : A_i(t) \leq 1\}$. Give every remaining selected mode the artificial deadline r . Sort these $r - 2$ modes by nondecreasing deadline, resolving ties arbitrarily, and assign them to the unit slots with start times

$$0, 1, \dots, j - 1, \quad j + 2, j + 3, \dots, r - 1.$$

We verify that each mode starts by its deadline. If the ℓ th sorted mode precedes q , it starts at $\ell - 1$. A deadline smaller than this would imply that all first ℓ modes have allocation strictly greater than one at $\ell - 1$, contradicting total allocation $\ell - 1$. If it follows q , it starts at $\ell + 1$, with $\ell \geq j + 1$. A deadline smaller than $\ell + 1$ would give allocation greater than one to the first ℓ modes there. Also $d_q \leq j + 2 \leq \ell + 1$, so q has allocation at least one there. Their combined allocation would exceed $\ell + 1$. An artificial deadline r cannot be violated by a slot start.

For a singly used mode, $W_i - A_i \leq 1$ because its total service is one. Before activation, $A_i - W_i \leq 1$ by the deadline property; after activation, its largest positive discrepancy is at r and is at most $A_i(r) - 1 \leq 1$. For q , the allocation at its block start is at most one and at its block end is at least one. Thus its positive discrepancy at the start and its negative discrepancy at the end are at most one. Its final positive discrepancy is $A_q(r) - 2 \leq 1$. Endpoint monotonicity controls the intervening times. Every omitted mode has allocation at most one throughout the prefix. The new prefix therefore has full error at most one. Since its counts at r are unchanged, attaching the original suffix preserves every later discrepancy. The two q slots are adjacent. $\square$

Proof of Theorem 3.1. Extend the input from T to $(s + 2)E$, when necessary, by assigning all added activity to any mode. An originally heavy mode remains heavy. Scale time and all allocations by E , giving the integer horizon $M = s + 2$. Lemmas 3.2 and 3.3 produce an error-one word with an adjacent pair. It has at most $M - 2 = s$ switches. Scale back and restrict to $[0, T]$; neither operation increases the switch count, and the error is at most E . $\square$

The reordered word need not preserve the stronger floor/ceiling bounds in (18); the conclusion needed is error at most one. Nor does the proof prescribe the repeated mode: the first repeated mode can differ from the heavy mode used in the flow objective.

Theorem 3.4 (Exact one-sided reduction). *For every $1 \leq k < n$,*

$$F_{n,k-1}(T) = \max \left\{ \frac{T}{k+1}, G_{n,k}^-(T) \right\}. \quad (20)$$

Repeated modes are allowed in both optimizations.

Proof. Set $E = \max\{T/(k+1), G_{n,k}^-(T)\} > 0$. For an input with a mass greater than E , Theorem 3.1 applies with $s = k-1$. Otherwise every mass is at most E . Choose a one-sided minimizing schedule, which exists by Proposition 1.2; its negative discrepancy is at most $G_{n,k}^-(T) \leq E$. Its positive discrepancy is bounded by $A_i(t) \leq m_i \leq E$, regardless of the schedule. This proves the upper bound. Full error dominates one-sided error for every input and schedule, so $F_{n,k-1} \geq G_{n,k}^-$. For the remaining lower bound, take $k+1$ successive distinct pure modes, each for time $T/(k+1)$. Every schedule with at most k blocks omits one of these modes and incurs its full terminal mass as error. The assumption $k < n$ permits this input. $\square$

4 Analytic reach bounds with two and three distinct modes

Distinct-mode schedules are a useful construction for the one-sided problem, although distinctness is never imposed on the competing optimum. For a threshold $E > 0$, a horizon $L > 0$, and an unused mode i , define

$$H_i(t) = t - A_i(t), \quad \Phi_i(b) = \max\{t \in [0, L] : H_i(t) \leq b + E\}, \quad 0 \leq b \leq L. \quad (21)$$

The functions H_i are continuous and nondecreasing, with $\sum_i H_i(t) = (n-1)t$. The feasible set is closed, nonempty, and contains b ; hence $\Phi_i(b) \geq b$. Activating unused mode i from b to t gives endpoint negative discrepancy $t - b - A_i(t) = H_i(t) - b$. Its maximum negative discrepancy occurs at the end of that block. Consequently a composition of reach maps along distinct modes constructs a schedule with one-sided error at most E . Each map is nondecreasing in b , so using the latest feasible endpoint at each step attains the greatest reach for that word.

We will repeatedly use the exact boundary facts

$$\Phi_i(b) < L \implies H_i(\Phi_i(b)) = b + E, \quad \Phi_i(b) < L \implies H_i(L) > b + E. \quad (22)$$

The first follows by continuity and maximality, and the second because L is then infeasible. Both remain valid with flat portions of H_i ; choosing an arbitrary threshold root instead of the latest endpoint would not suffice for all comparisons below. Put

$$r_n = \frac{n}{n-1}, \quad B_j(n, E) = nE(r_n^j - 1), \quad B_0(n, E) = 0. \quad (23)$$

When n, E are fixed, write $B_j = B_j(n, E)$. These values satisfy $B_j = r_n(B_{j-1} + E)$.

Lemma 4.1 (Two distinct blocks). *For $n \geq 3$, two distinct modes suffice to reach $\min\{T, B_2\}$ with one-sided error at most E . A single mode suffices to reach $\min\{T, B_1\}$ for every $n \geq 2$.*

Proof. Define reaches on the target horizon L . If a candidate already reaches L , there is nothing to prove. Otherwise the first reaches $R_i = \Phi_i(0)$ are uncapped, so $A_i(R_i) = R_i - E$. For the single-block assertion, $x = \max_i R_i < L$ implies $(n-1)x = \sum_i H_i(x) \geq nE$, so $x \geq B_1$, a contradiction when $L \leq B_1$.

For two blocks, assume every ordered distinct pair has reach below L , and let $M = \max_{p \neq q} \Phi_q(R_p) < L$. Let $x \geq y$ be the two largest first reaches and choose p attaining x . At

time y , $A_p(y) \leq x - E$ and $A_i(y) \leq y - E$ for every $i \neq p$. Summing gives $x + (n-2)y \geq nE$. The identity

$$(n-1)x + y = \frac{n}{n-1}[x + (n-2)y] + \frac{n^2 - 3n + 1}{n-1}(x - y) \geq \frac{n^2 E}{n-1} \quad (24)$$

uses $n \geq 3$. For each i , let $x_i = \max_{j \neq i} R_j$. Every pair ending in i reaches no further than M , so (22) and monotonicity give $A_i(M) \leq M - E - x_i$. Sum over i and note $\sum_i x_i = (n-1)x + y$. It follows that

$$(n-1)M \geq nE + (n-1)x + y, \quad M \geq \frac{nE(2n-1)}{(n-1)^2} = B_2. \quad (25)$$

This contradicts $M < L \leq B_2$ and proves the claim. $\square$

Theorem 4.2 (Three distinct blocks). *For $n \geq 4$, three distinct modes suffice to reach $\min\{T, B_3\}$ with one-sided error at most E .*

Proof. Take $L = \min\{T, B_3\}$ and suppose no sequence of at most three distinct modes reaches L . In particular every first and pair reach is uncapped. Retain R_i, M, x, y from Lemma 4.1, and let z be the third largest first reach. At time z , the top two modes have allocation at most $x - E$ and $y - E$, and the other modes have allocation at most $z - E$. Thus $x + y + (n-3)z \geq nE$. Writing $u = x + y \geq 2z$ gives

$$\begin{aligned} (n-1)u + 2z &= \frac{2n}{n-1}[u + (n-3)z] + \frac{n^2 - 4n + 1}{n-1}(u - 2z) \\ &\geq \frac{2n^2 E}{n-1}, \end{aligned} \quad (26)$$

where the last coefficient is positive for $n \geq 4$.

Let M_i be the largest pair reach excluding mode i , and let $x_i \geq y_i$ be the largest two first reaches after that exclusion. For each $a \neq i$, all pairs available without i and ending in a give

$$A_a(M_i) \leq M_i - E - \max_{b \notin \{i, a\}} R_b.$$

The maxima on the right sum over $a \neq i$ to $(n-2)x_i + y_i$. Summing and using $\sum_a A_a(M_i) = M_i$ yields

$$A_i(M_i) \geq (n-1)E + (n-2)x_i + y_i - (n-2)M_i.$$

On the other hand, $M_i \leq M$ and $A_i(M) \leq M - E - x_i$. Monotonicity therefore implies

$$(n-2)M_i \geq nE + (n-1)x_i + y_i - M. \quad (27)$$

Choose a pair (p, q) attaining M . The same pair is available in every exclusion except possibly p, q , so $M_i = M$ outside this pair. Also $x_p + x_q \geq x + y$ and $y_p, y_q \geq z$. Summing (27) for p, q gives

$$\sum_i M_i \geq \left(n - 2 - \frac{2}{n-2}\right) M + \frac{2nE + (n-1)(x+y) + 2z}{n-2}.$$

The coefficient of M is positive for $n \geq 4$. Substituting $M \geq B_2$ from (25) and (26) proves

$$\sum_i M_i \geq nB_2. \quad (28)$$

For completeness, the substituted expression equals nB_2 because $(n-1)B_2 = nE + n^2E/(n-1)$.

Appending mode i to a maximizing pair excluding it fails to reach L by assumption. The second part of (22) gives $H_i(L) > M_i + E$. Sum these strict inequalities and use (28):

$$(n-1)L > \sum_i M_i + nE \geq nB_2 + nE = (n-1)B_3.$$

This contradicts $L \leq B_3$. $\square$

This proof is constructive: compute the first reaches and the ordered-pair reach matrix, then append each possible final mode to a maximizing pair excluding it. A global maximizing pair supplies all but at most two of these excluded maxima; the remaining two require additional scans of the matrix. The construction therefore uses $n(n-1)$ pair reach evaluations, followed by n final reach evaluations and $O(n^2)$ comparisons. It replaces an earlier proof for $n = 4$ based on thirty event-order linear programs. Those exact certificates are included as an independent special-case check in the supplement, but are not premises of Theorem 4.2.

5 Four distinct blocks: an exact certificate proof

The next reach bound has the same uniform-input constant, but its proof uses a finite family of exact linear identities. We give the complete necessary-condition relaxation and its symmetry reduction before applying the certificates. The computer-assisted part establishes only the weighted pair inequality below; the passage from that inequality to a four-block schedule is analytic. No time discretization of the input is involved.

Theorem 5.1 (Four distinct blocks, computer-assisted). *For $n \geq 5$, four distinct modes suffice to reach $\min\{T, B_4(n, E)\}$ with one-sided error at most E .*

5.1 The weighted pair inequality

Use the reach maps (21) on $[0, L]$. Let $R_i = \Phi_i(0)$ and define the pair maxima

$$\begin{aligned} M &= \max_{a \neq b} \Phi_b(R_a), \\ P_i &= \max_{a, b \notin \{i\}, a \neq b} \Phi_b(R_a), \\ Q_{ij} = Q_{ji} &= \max_{a, b \notin \{i, j\}, a \neq b} \Phi_b(R_a), \quad i \neq j. \end{aligned} \tag{29}$$

These sets of pairs are nonempty. Assume $M < L$, so all pair and first reaches are uncapped. For a three-element subset $S \subset [n]$, write

$$\mathcal{H}_S = \sum_{i \in S} \left(P_i + \sum_{j \neq i} Q_{ij} \right). \tag{30}$$

An excluded pair wholly in S is counted twice in this sum.

Lemma 5.2 (Weighted pair inequality, computer-assisted). *For $n \geq 5$, $M < L$, any three-element set S , and any distinguished mode z ,*

$$\mathcal{H}_S \geq \frac{3n^2}{n-1}E + \frac{3n^2}{(n-1)^2} \sum_{i \neq z} R_i. \tag{31}$$

In particular,

$$\mathcal{H}_S \geq 3nB_2(n, E). \tag{32}$$

To see the implication between the two displayed inequalities, choose z with smallest first reach. Then

$$R_z = \sum_i A_i(R_z) \leq \sum_i A_i(R_i) = \sum_i (R_i - E), \quad \sum_{i \neq z} R_i \geq nE.$$

Substitution in (31) gives (32), since $B_2 = nE(2n-1)/(n-1)^2$. The stronger statement is what the certificates prove, and does not assume that z is smallest.

5.2 The complete necessary-condition linear program

Scale time by E , so that $E = 1$. For this subsection and its implementation only, label modes $0, \dots, n-1$ and set $S = \{0, 1, 2\}$. The variables are nonnegative R_i , and event times t_v and allocations $a_{v,i}$ for

$$v \in \mathcal{V} = \{M\} \cup \{P_i : i \in S\} \cup \{Q_{ij} : i < j, \{i, j\} \cap S \neq \emptyset\}.$$

There are $1 + 3 + [3(n-3) + 3] = 3n - 2$ events. For event v , let D_v be its excluded set: $\emptyset$, $\{i\}$, or $\{i, j\}$ as appropriate. Choose an unordered pair $P = \{p, q\}$ attaining the global pair maximum M . The relaxation consists of exactly the following constraints:

$$\sum_{i=0}^{n-1} a_{v,i} = t_v \quad (v \in \mathcal{V}), \quad (33)$$

$$R_j - t_v + a_{v,k} \leq -1 \quad (v \in \mathcal{V}; j, k \notin D_v, j \neq k), \quad (34)$$

$$a_{Q_{ij},k} \leq a_{P_i,k} \quad (i \in S, j \neq i; 0 \leq k < n), \quad (35)$$

$$a_{v,k} \leq a_{M,k} \quad (v \in \mathcal{V} \setminus \{M\}; 0 \leq k < n), \quad (36)$$

$$a_{M,k} \leq a_{Q_{ij},k} \quad (\{i, j\} \cap P = \emptyset, Q_{ij} \in \mathcal{V}; 0 \leq k < n), \quad (37)$$

$$a_{M,k} \leq a_{P_i,k} \quad (i \in S \setminus P; 0 \leq k < n). \quad (38)$$

All variable nonnegativity constraints are imposed as well. The objective is

$$C = (n-1)^2 \left(\sum_{i \in S} t_{P_i} + \sum_{\substack{i < j \\ \{i, j\} \cap S \neq \emptyset}} |\{i, j\} \cap S| t_{Q_{ij}} \right) - 3n^2 \sum_{i \neq z} R_i. \quad (39)$$

We will certify $C \geq 3n^2(n-1)$.

Every input in the uncapped case gives a feasible point: set t_v to its pair reach and $a_{v,i} = A_i(t_v)$. Equation (33) expresses total allocation. For (34), the event reach is at least $\Phi_k(R_j)$ and this endpoint is uncapped. Hence $H_k(t_v) \geq R_j + 1$. Excluding more modes cannot increase a pair maximum, which gives (35) and (36) by allocation monotonicity. If an exclusion avoids the maximizing pair P , that pair remains available and its event time equals M . The reverse allocation inequalities (37)–(38) express these equalities. Together with (33), the two allocation orders also force the event times to be equal.

There are no additional constraints: in particular we impose neither first-root equations nor an ordering of the R_i , and no chronological ordering between unrelated events. The relaxation can therefore include nonphysical points. A lower bound valid on this larger feasible set is nevertheless valid for every control. No converse representation claim is used.

P	Representatives for z	Range
$\{0, 1\}$	$0, 2, 3$	$n \geq 5$
$\{0, 3\}$	$0, 1, 3, 4$	$n \geq 5$
$\{3, 4\}$	$0, 3$	$n \geq 5$
$\{3, 4\}$	5	$n \geq 6$

Table 1: All ten types of (S, P, z) after relabeling $S = \{0, 1, 2\}$.

5.3 Exhaustive symmetry reduction

Permutations preserving S carry a maximizing pair to one of three representatives, according to its intersection size with S . Permutations preserving both S and this pair then place z in the representatives shown in Table 1. For example, when $P = \{0, 3\}$, the four possible positions of z are the singleton $S \cap P$, one of the two elements of $S \setminus P$, the singleton $P \setminus S$, and an element outside $S \cup P$. These are exactly the four displayed cases. The same partition argument proves the other two rows. Ties between maximizing pairs do not matter: any maximizing pair may be chosen.

For one type, keep the indices in $K = S \cup P \cup \{z\}$ individually labeled and let the remaining indices be interchangeable. Averaging a feasible point over all permutations of those remaining indices preserves every constraint and the objective. Thus restricting variables to be constant on permutation orbits preserves the infimum of the relaxation. This assertion needs no optimizer or boundedness assumption: every feasible point has an invariant feasible average with the same value.

Here is an explicit orbit rule sufficient to reconstruct the quotient. A variable is identified by (R, i) , (t, v) , or (a, v, i) . Retain each index in K verbatim. Replace each other index, in order of first occurrence in this identifier, by $b_0, b_1, \dots$, reusing its symbol on subsequent occurrences. A Q event contains at most one interchangeable index, because its excluded pair meets S . Thus this rule preserves exactly the equalities and inequalities of the unspecified indices, without an orientation ambiguity for its unordered pair. In particular, allocation to the interchangeable index excluded by a Q event and allocation to a different interchangeable index remain separate variables.

Replace variables in each row of (33)–(38) by these orbit labels, add coefficients that now refer to the same variable, and delete duplicate rows. In the objective, add all multiplicities of the same variable orbit. This defines the quotient matrices and objective uniquely, independently of any numerical solver. The bundled checker implements precisely this construction. Its ordering enumerates R_i first, then M , the Q_{ij} in lexicographic order, and the three P_i ; it retains the first occurrence of each variable orbit and each distinct row. The certificate coordinates refer to that ordering.

At most six indices lie in K . A pair-constraint row involves at most three interchangeable indices: one from its event and its two available modes. An allocation-order row involves at most two. Thus all variable and row patterns are already present once $n \geq 9$. Additional interchangeable indices repeat those patterns. They affect only multiplicities in mass equations and the objective. If $a = |K|$, a mass row has $n - a$ identical allocation terms, or one separately labeled term and $n - a - 1$ others when an interchangeable index is excluded by the event. Hence its coefficients are affine functions of n . Objective orbit multiplicities are constant or $n - a$, multiplied by $(n - 1)^2$ for event times or by $-3n^2$ for first reaches. All objective coefficients are consequently integer polynomials of degree at most three. The inequality matrix and right-hand sides are constant for $n \geq 9$.

For $n \geq 9$, the quotient has between 57 and 162 variables, 151 and 826 inequality rows,

and 10 and 19 equality rows, depending on the type. For the symbolic check, the exact affine mass coefficients and exact affine objective multiplicities are reconstructed from the quotient at $n = 9, 10$. This is reconstruction of the established affine counts, not interpolation of an unknown function. All individually labeled indices are at most 5, and three interchangeable indices suffice for every row pattern. First-occurrence ordering of patterns is therefore stable when further indices are added.

5.4 Exact dual identities and the verification boundary

Write the quotient as $Ax \leq b$, $B(n)x = d$, $x \geq 0$, with objective $c(n)^T x = C$. In this notation b has entries zero or minus one and $d = 0$. For a finite dimension, a certificate supplies a positive integer denominator D and integer vectors $Y \leq 0$, Z satisfying

$$Dc = A^T Y + B^T Z, \quad b^T Y + d^T Z = 3n^2(n-1)D. \quad (40)$$

For every feasible x , multiplying its inequalities by the nonpositive entries of Y reverses their direction. Therefore

$$DC = Y^T Ax + Z^T Bx \geq Y^T b + Z^T d = 3n^2(n-1)D.$$

No dual optimality assertion is required. The certificates have zero coefficient residual, so nonnegative-variable residual multipliers are unnecessary.

For $5 \leq n \leq 22$, the data contain all nine types at $n = 5$ and all ten at each $n = 6, \dots, 22$: a total of $9 + 17 \cdot 10 = 179$ finite certificates. The checker verifies the case identifiers, signs, integrality, every coordinate of (40), and the stated lower bound.

For each type, a further certificate gives integer polynomials $D(n)$, $D_0(n)$, $Y(n)$, and $Z(n)$ satisfying

$$D(n) = (n-1)^2 D_0(n), \quad (41)$$

$$D_0(n)c(n) = A^T Y(n) + B(n)^T Z(n), \quad (42)$$

$$b^T Y(n) + d^T Z(n) = 3n^2(n-1)D_0(n). \quad (43)$$

The checker verifies these coefficient by coefficient, using integer polynomial addition and multiplication. It expands $D(23+t)$ and each entry of $-Y(23+t)$. The former has positive constant coefficient and nonnegative remaining coefficients; the latter have only nonnegative coefficients. Consequently $D(n) > 0$ and $Y(n) \leq 0$ for every real $n \geq 23$, and (41) gives $D_0(n) > 0$ there. Dividing (42)–(43) by $D_0(n)$ gives the same dual bound for every integer $n \geq 23$. These checks cover all dimensions $n \geq 5$ and prove (31).

The supplement's `verification/reference/` directory contains the complete data in `certificates_general_four_block.json` and the checker `verify_general_four_block.py`. Run, from the paper directory,

```
python verification/reference/verify_general_four_block.py
```

The checker requires only the Python standard library and uses integer arithmetic throughout. It reads certificate arrays as data, not executable expressions. Numerical optimization and symbolic algebra were used to discover the certificates; neither a numerical solution, a floating-point tolerance, nor an optimization status is trusted by the checker. Its successful output verifies all 179 finite cases and all ten polynomial cases. A provenance manifest records SHA-256 hashes of the bundled original artifacts. The proof's computational dependency is this finite data and its exact verification, together with the mathematical reduction and orbit coverage above. It is explicitly a computer-assisted proof.

5.5 From weighted pairs to four distinct blocks

Proof of Theorem 5.1. Put $L = \min\{T, B_4\}$ and assume no sequence of at most four distinct modes reaches L . All first, pair, and triple reaches are then uncapped. Let G be the maximum three-distinct-mode reach and N_i the corresponding maximum excluding i . Theorem 4.2 implies $G \geq B_3$: if $L \leq B_3$, that theorem already contradicts the assumed failure.

For each $k \neq i$, appending k to a pair attaining Q_{ik} gives a triple excluding i and ending no later than N_i . Hence $A_k(N_i) \leq N_i - E - Q_{ik}$. Summing over $k \neq i$ gives

$$A_i(N_i) \geq (n-1)E + \sum_{k \neq i} Q_{ik} - (n-2)N_i.$$

Appending i to a pair attaining P_i similarly gives $A_i(G) \leq G - E - P_i$. As $N_i \leq G$, allocation monotonicity implies

$$(n-2)N_i \geq nE + P_i + \sum_{k \neq i} Q_{ik} - G. \quad (44)$$

Choose a triple attaining G and let S be its mode set. Then $N_i = G$ outside S . Sum (44) over its three elements:

$$\sum_i N_i \geq \left(n - 3 - \frac{3}{n-2}\right)G + \frac{3nE + \mathcal{H}_S}{n-2}.$$

The coefficient of G is positive for $n \geq 5$. Applying $G \geq B_3$ and Lemma 5.2 yields

$$\sum_i N_i \geq nB_3, \quad (45)$$

because $n(B_2 + E) = (n-1)B_3$.

Finally, appending i to a maximizing triple excluding it uses four distinct modes. Its failure implies $H_i(L) > N_i + E$. Summing gives

$$(n-1)L > \sum_i N_i + nE \geq nB_3 + nE = (n-1)B_4,$$

contradicting $L \leq B_4$. The strict inequality follows from latest endpoints even when H_i has flat portions. Endpoint monotonicity makes the endpoint constraints control the entire one-sided error trajectory; unselected modes have $W_i - A_i = -A_i \leq 0$. $\square$

An earlier five-mode proof used a separate weighted-pair event relaxation. Its verifier and certificate data are retained in the supplement as an independent check of the smallest case. The all-dimension certificates above subsume that positive result. Neither the five-mode checker nor the thirty-case three-block checker is needed to establish the theorem as presented here.

6 Exact minimax values through three switches

The analytic reach bounds and the four-block certificate theorem determine both the one-sided minimax and the full minimax. A common formulation also preserves the stronger conclusion available under equal terminal masses.

Theorem 6.1 (One-sided minimax through four blocks). *For $1 \leq k \leq 4$ and $n \geq k+1$,*

$$G_{n,k}^-(T) = \frac{T}{n(r_n^k - 1)}. \quad (46)$$

The constant uniform input attains this value. For $k = 4$ the upper bound uses the computer-assisted Theorem 5.1; for $k \leq 3$ its proof is analytic.

Proof. Use the single-, two-, three-, or four-block reach result, respectively, at $E = T/[n(r_n^k - 1)]$. It reaches T and proves the upper bound with distinct modes. For the uniform input $A_i(t) = t/n$, let an arbitrary schedule, allowing repetitions, have one-sided error E and block endpoints t_j . At the end of block j , the active mode's occupation is at least $t_j - t_{j-1}$, so

$$t_j - t_{j-1} - t_j/n \leq E, \quad t_j \leq r_n(t_{j-1} + E).$$

Iteration yields $T \leq nE(r_n^k - 1)$. Shorter schedules can be padded with zero-length blocks, or the recurrence can be iterated further using its monotonicity. This proves the matching lower bound even when modes repeat. $\square$

Theorem 6.2 (Full minimax through three switches). *For $0 \leq s \leq 3$ and $n \geq s + 2$,*

$$F_{n,s}(T) = T \max \left\{ \frac{1}{s+2}, \frac{1}{n(r_n^{s+1} - 1)} \right\}. \quad (47)$$

In particular,

$$F_{n,2}(T) = T \max \left\{ \frac{1}{4}, \frac{(n-1)^3}{n(3n^2 - 3n + 1)} \right\}, \quad n \geq 4, \quad (48)$$

$$F_{n,3}(T) = T \max \left\{ \frac{1}{5}, \frac{(n-1)^4}{n(4n^3 - 6n^2 + 4n - 1)} \right\}, \quad n \geq 5. \quad (49)$$

The three-switch upper bound is computer-assisted; the other cases are analytic.

Proof. Combine Theorems 3.4 and 6.1. The two matching input types are the $s+2$ successive distinct pure modes of equal duration and the constant uniform input. Theorem 2.1 gives the latter's full error, whose additional T/n term is bounded by $T/(s+2)$ in this range. $\square$

Corollary 6.3 (Equal terminal masses). *Let $1 \leq k \leq 4$ and $n \geq k + 1$. Over inputs with $m_i = T/n$ for every mode, the exact worst-case full error with at most k blocks is*

$$\max_{\substack{A \in \mathcal{A}_n(T) \\ A_i(T) = T/n \ (i \in [n])}} \text{OPT}_{k-1}(A) = T \max \left\{ \frac{1}{n}, \frac{1}{n(r_n^k - 1)} \right\}. \quad (50)$$

The constant uniform input is an extremizer in this restricted class.

Proof. The reach theorem controls $W - A$ at the second threshold in (50). Independently, every positive discrepancy is bounded by $A_i(t) \leq m_i = T/n$. Combining these two bounds proves the upper bound. The uniform input belongs to the class and attains the displayed value by Theorem 2.1. $\square$

The equal-mass conclusion is stronger than merely applying the unrestricted minimax bound to these inputs. For two switches it gives $T/4$ at $n = 4$ and the uniform recurrence value for all $n \geq 5$. Thus at $n = 5, 6, 7$ the restricted worst case is strictly smaller than the unrestricted value $T/4$. For three switches it gives T/n at $n = 5, 6$ and the uniform recurrence value for $n \geq 7$; it is strictly smaller than $T/5$ for $n = 6, \dots, 11$. The common corollary derives the equal-mass result for four blocks as well as the earlier analytic three-block result without needing a separate construction for each mass-constrained class.

6.1 Plateau transitions and mode-count asymptotics

For two switches, the uniform coefficient in (48) exceeds $1/4$ exactly when

$$p_2(n) = n^3 - 9n^2 + 11n - 4 > 0.$$

Direct substitution gives negative values at $n = 4, 5, 6, 7$. At $n = 8 + h$,

$$p_2(8 + h) = h^3 + 15h^2 + 59h + 20 > 0 \quad (h \geq 0).$$

Thus $F_{n,2}(T) = T/4$ for $4 \leq n \leq 7$, and uniform controls are worst-case for all $n \geq 8$.

For three switches, the corresponding sign polynomial is

$$p_3(n) = n^4 - 14n^3 + 26n^2 - 19n + 5.$$

It is negative at $n = 5, \dots, 11$, whereas

$$p_3(12 + h) = h^4 + 34h^3 + 386h^2 + 1469h + 65 > 0 \quad (h \geq 0).$$

Hence $F_{n,3}(T) = T/5$ for $5 \leq n \leq 11$, and uniform controls are worst-case for $n \geq 12$. Expanding the exact rational expressions gives

$$F_{n,2}(T) = \frac{T}{3} - \frac{2T}{3n} + \frac{2T}{9n^2} + O(Tn^{-3}), \quad (51)$$

$$F_{n,3}(T) = \frac{T}{4} - \frac{5T}{8n} + \frac{5T}{16n^2} + O(Tn^{-3}). \quad (52)$$

These are expansions of the full minimax, since the uniform term dominates for all sufficiently large n , rather than lower-bound expansions alone.

7 Structural limits of the constructions

The hypotheses and choices in the preceding arguments matter. We give exact counterexamples to three tempting extensions. In the first example, the failure extends beyond distinct-mode schedules: allowing a repeated mode still does not restore uniform extremality at the boundary $k = n$. This strengthens what a failure of a distinct-mode construction alone would imply.

7.1 A three-mode obstruction with an exact instance optimum

Proposition 7.1 (Failure at $k = n = 3$). *There is an input $A \in \mathcal{A}_3(57/8)$ with*

$$\text{OPT}_3^-(A) = \frac{18673}{18396} > 1, \quad (53)$$

although the constant uniform input on that horizon has one-sided optimum exactly one with three blocks. An optimal word is $(0, 2, 1)$, with switching times $8341/4599$ and $17639/4599$. Moreover, every three-distinct-mode reach composition at threshold one stops at $971/146 < 57/8$ for this input. In particular, $G_{3,3}^-(57/8) \geq 18673/18396 > 1$.

Proof. For this example label the modes $0, 1, 2$. Define their cumulative allocations by linear interpolation through Table 2, dividing every entry by 146. After its last knot $t_* = 971/146$, extend with rates $(1/3, 1/3, 1/3)$ to $L = 57/8$. Every allocation increment is nonnegative and the three increments sum to its time increment. Each individual slope is at most $3/4$, so this is an admissible control and every $H_i(t) = t - A_i(t)$ is strictly increasing.

$146t$	$146A_0(t)$	$146A_1(t)$	$146A_2(t)$
0	0	0	0
146	98	48	0
194	110	48	36
256	110	78	68
408	224	116	68
516	224	178	114
580	240	178	162
971	417	309	245

Table 2: Rational cumulative-allocation knots for the three-mode example. Every entry, including time, is divided by 146.

At threshold $E = 1$ the first reaches are

$$(R_0, R_1, R_2) = (128/73, 97/73, 1).$$

Both orders of the other two modes after excluding i have the same pair reach, respectively

$$(M_0, M_1, M_2) = (204/73, 258/73, 290/73).$$

These identities follow by substituting the relevant knots into $H_i(R_i) = 1$ and $H_k(M_i) = R_j + 1$ for distinct i, j, k . At the last knot, $H_i(t_*) = M_i + 1$ for each i . Strict increase proves that all six three-distinct-mode compositions end exactly at t_* .

We first bound distinct-mode schedules at thresholds slightly above one. Write

$$\delta_* = \frac{277}{18396}, \quad E_* = 1 + \delta_* = \frac{18673}{18396}, \quad L - t_* = \frac{63}{2}\delta_*.$$

For $y \geq 0$, define the capped inverse $\psi_i(y) = \max\{t \in [0, L] : H_i(t) \leq y\}$, so the threshold- E reach map is $\Phi_i^E(b) = \psi_i(b + E)$. The slope bound gives $H_i(t) - H_i(u) \geq (t - u)/4$ for $u \leq t$. Consequently ψ_i is nondecreasing and 4-Lipschitz. Indeed, if $\psi_i(y) < \psi_i(z)$, the lower endpoint is uncapped and satisfies $H_i(\psi_i(y)) = y$, whereas $H_i(\psi_i(z)) \leq z$, even if the upper endpoint is L . This proves $\psi_i(z) - \psi_i(y) \leq 4(z - y)$ without excluding caps.

For $E = 1 + \delta$ with $0 \leq \delta \leq \delta_*$, the first reach $R_p(E) = \Phi_p^E(0)$ is at most $R_p + 4\delta$. A pair using the two modes other than i , in either order, therefore reaches at most $M_i + 4(4\delta + \delta) = M_i + 20\delta$. On the uniform suffix,

$$H_i(L) = M_i + 1 + \frac{2}{3}(L - t_*) = M_i + 1 + 21\delta_*. \quad (54)$$

For the last mode i to reach L , its previous endpoint b must satisfy $H_i(L) \leq b + E \leq M_i + 1 + 21\delta$. This is impossible when $\delta < \delta_*$. Equivalently, every distinct triple reaches at most $\min\{L, t_* + (63/2)\delta\}$, since its last inverse starts at t_* and has slope $3/2$ on the uniform suffix.

The word $(0, 2, 1)$ attains all these bounds at $\delta = \delta_*$. In fact, H_0 has slope $1/4$ on $[256/146, 408/146]$, starting at R_0 ; H_2 has slope $1/4$ on $[516/146, 580/146]$, starting at M_1 . The times

$$u_* = R_0 + 4\delta_* = \frac{8341}{4599}, \quad v_* = M_1 + 20\delta_* = \frac{17639}{4599} \quad (55)$$

lie in these respective intervals. Thus $H_0(u_*) = E_*$ and $H_2(v_*) = u_* + E_*$. Equation (54) gives $H_1(L) = v_* + E_*$. Each active mode has negative discrepancy exactly E_* at its block end. Endpoint monotonicity proves that this schedule has one-sided error E_* .

We must still exclude repeated-mode schedules below this value. The terminal masses are

$$(m_0, m_1, m_2) = \frac{1}{1752}(5281, 3985, 3217), \quad m_0 > 2E_*, \quad m_1 > 2E_*. \quad (56)$$

For a schedule with active-mode set J and one-sided error at most E_* , its terminal inequalities imply

$$L = \sum_{i \in J} W_i(L) \leq \sum_{i \in J} (m_i + E_*), \quad \sum_{i \notin J} m_i \leq |J|E_*.$$

If at most two modes are used, neither mode 0 nor mode 1 can be omitted. The only possible active set is $\{0, 1\}$. Every schedule on this set with at most three blocks has the form p, q, p , with $(p, q) = (0, 1)$ or $(1, 0)$, allowing zero-length blocks to include shorter schedules.

Let $u \leq v$ be its first two endpoints. Since q has not been used before its middle block, their constraints give $u \leq R_p(E_*)$ and $v \leq \Phi_q^{E_*}(u) \leq M_2 + 20\delta_*$. The latter upper bound is below L . The allocation slope bound $3/4$ therefore yields

$$v - u \leq A_q(v) + E_* \leq A_q(M_2) + 15\delta_* + 1 + \delta_* = M_2 - R_p + 16\delta_*. \quad (57)$$

Here $A_q(M_2) + 1 = M_2 - R_p$ follows from the threshold-one pair identity. The final occupation of p is $L - (v - u)$, so its terminal constraint requires $v - u \geq L - m_p - 1 - \delta_*$. For $p = 0$ and $p = 1$, respectively, the necessary lower bound minus the upper bound in (57) is

$$\frac{781}{876} - 17\delta_* = \frac{2923}{4599} > 0, \quad \frac{1057}{876} - 17\delta_* = \frac{4372}{4599} > 0.$$

Thus no repeated-mode schedule is feasible even at E_* . Feasibility is monotone in the error threshold, so none is feasible below E_* . The distinct-mode argument excludes every threshold in $[1, E_*)$; infeasibility at one also excludes smaller thresholds. Together with the matching schedule, this proves (53).

Finally, the uniform input has the distinct-mode schedule with endpoints $3/2, 15/4, 57/8$; each active mode has negative discrepancy one at the end of its block. Conversely, the block-end recurrence $t_j \leq (3/2)(t_{j-1} + E)$, valid even with repeated modes, gives $57/8 \leq 3E((3/2)^3 - 1) = (57/8)E$. Thus its one-sided optimum is exactly one, as asserted. $\square$

By scaling, the example gives

$$G_{3,3}^-(T) \geq \frac{37346}{262143}T > \frac{8}{57}T,$$

where the last coefficient is the uniform-input one-sided optimum per unit horizon. This is a lower bound on the minimax, not a determination of its exact value. The supplement checks the knot slopes, all six threshold-one reach compositions, inverse-sensitivity arithmetic, matching schedule, terminal masses, and both middle-service contradictions using rational arithmetic. It also independently rules out every three-block word at threshold one by exact switching-time polygon feasibility. This counterexample does not concern the spare-mode range $k < n$ of Theorems 3.4 and 6.1, nor does it have equal terminal masses.

7.2 A specified heavy mode need not admit an adjacent pair

Theorem 3.1 chooses a repeat; it does not promise to repeat any prescribed heavy mode. On four unit intervals, consider the interval rates

mode	1	2	3	4	total
0	7/15	0	0	0	7/15
1	2/15	1/2	0	4/5	43/30
2	2/5	1/2	1	1/5	21/10

Mode 1 is heavy at threshold one. If an error-one unit-slot word selected it in slots 1–2 or 2–3, its occupation at the pair’s end would be at least two, but its allocation is only $19/30$. This gives negative error at least $41/30 > 1$. If its pair occupied slots 3–4, mode 2 would need at least two selections in slots 1–2, because its terminal mass is $21/10 > 2$. Yet two such selections give error $2 - 9/10 = 11/10 > 1$ at prefix 2. No location of the prescribed pair works. The words $(0, 2, 2, 1)$ and $(0, 1, 2, 2)$ both have error at most one, consistently with the existential theorem. This statement concerns rounding on the fixed unit slots; it does not impose a restriction on arbitrary continuous block lengths.

7.3 The pair position cannot be fixed by the first unit-mass prefix

Even if a largest-mass mode is chosen, the following rule can fail: force its two adjacent slots to end at the first integer prefix where its allocation reaches one. On five unit intervals take

mode	1	2	3	4	5	total
0	$3/5$	0	$1/10$	$2/5$	1	$21/10$
1	$3/10$	$3/5$	$11/20$	$3/5$	0	$41/20$
2	$1/10$	$2/5$	$7/20$	0	0	$17/20$

Mode 0 has strictly largest mass and first reaches allocation at least one at prefix 4. Forcing it in slots 3–4 leaves mode 1 with no service there. At prefix 4, mode 1 already has allocation $41/20 > 2$, so an error-one word must have selected it in both slots 1–2. Its allocation at prefix 2 is only $9/10$, giving the impossible negative error $11/10$. In contrast, $(0, 1, 1, 0, 0)$ has error at most one and repeats the largest mode in slots 4–5. Thus both the selected mode and the pair position are meaningful choices in the prefix-reordering proof. These examples do not disprove the separate conjecture that some error-one word can always repeat a largest heavy mode adjacently; that strengthening is not used by any theorem here.

References

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